LEF MODEL · CHAPTER 14

Chapter 14: External Rotational Differentiation — Polar Contraction, the Thin Polar Crust and the Crater-Size Tests

Based on Paper 07 — The Poles: Why the Poles Are Crowded (Parts C.3–E: the geometry of contraction, the energy balance, the 4 tests and the 5 diameter-preservation tests)

01 · THE MYSTERY

The mystery: if the poles contracted by half — why is the crust beneath the poles thinner, and why did the craters themselves not shrink?

In Chapter 13 we saw that the young Moon passed from a flattened, elongated body ($a : b : c \approx 1.8 : 1.4 : 1$) to a spherical shape while its volume contracted, and that this process crowded the polar craters (1.20 ± 0.05 in the whole population and 1.53 ± 0.19 in craters of 60–100 km). But here **three key physical questions** at once arise which, if left unresolved, would bring down the whole model:

1. **The paradox of polar crustal thickness:** if the polar area shrank to about **0.55×** of its original area during the rounding, the material in the polar shell was compressed inward and the crust beneath the poles should have been *thicker* than the crust beneath the equator! In fact, the GRAIL measurements show the exact opposite: **the crust beneath the poles is thinner and denser than the crust beneath the equator!**

2. **The paradox of crater-size preservation:** if the surface at the poles contracted, why did the craters themselves not contract with it? Had the craters shrunk in diameter, craters near the counting threshold (20 km) would have fallen below it and wiped out the density excess completely!

3. **The engine of energy and pressure:** what physical engine drove the inward rounding after the corotation line was crossed — and what happened to the enormous pressure (2–12 GPa) that this rounding exerted on the rigid outer shell?

02 · AN IMAGE

An image: a bowl of honey that levels out, stiff tiles on a shrinking sheet, and "external rotational differentiation"

Three simple images solve the three riddles and unite them into one geodynamic process:

- **A bowl of honey that was tilted and set back level (the gravity engine, §C.4):** while the Moon was attached to the Earth and spinning fast (5.5 hours), the centrifugal force held the poles "energetically raised" by 120 km above the equator (like an energy hill from which light material slid toward the equator). The moment the centre of mass crossed the corotation line (15,818–16,399 km), the stretching at the base of the pear fell to zero and the density rose from 1,700 to 2,600 kg/m³. The Moon's self-gravity suddenly overcame the rotation — like a figure skater whose weight doubled all at once and who can no longer hold her arms out — and the pole at once turned from a "mountain top" into a "valley floor" into which all the material flows under its own weight!
- **Stiff ceramic tiles on a shrinking rubber sheet (preserving crater size, §C.7, §D.2):** the upper shell had already frozen and was stiff enough to carry the craters as **stiff, whole tiles** that keep their original diameter, while the elastic gaps between the craters closed and were absorbed into the structure of the crust!

- **"External rotational differentiation" (§C.8, §D.6):** why is the crust beneath the poles thinner today despite the contraction? Because during **the tens of millions of years of the flattened stage** (before the contraction began!), the fast rotation made the light material of the outer shell flow outward **from the poles toward the equator**, creating an *especially thin* polar shell. When the rounding stage came, this thin shell was compressed inward and became **a thin, dense polar crust** (still thinner than the thick equatorial crust!), while the craters that had landed evenly on the stretched shell were crowded toward the centres of the poles!

03 · SCIENCE AND MODEL

Science and model: computing the area contraction (0.55×) and the energy balance (1.5 × 10^{28} joules)

Parts C.3 and C.4 of Paper 07 quantify with mathematical precision both the change of area at every point on the Moon's face and the release of energy and internal pressure during the rounding:

1. **The absolute area contraction combined with the drop in volume (§C.3.1):** in passing from peak flattening ($\rho \approx 1,700$ kg/m³, equivalent radius 2,180 km, dimensions 5,585 × 4,628 × 3,191 km) to the spherical body at detachment ($\rho \approx 2,600$ kg/m³, spherical diameter 3,778 km, radius 1,890 km), the volume falls by a factor of **1.53**, the equivalent radius contracts by 13%, and the mean surface area falls to **0.75×**. But because of the triaxial change of shape (1.75 : 1.45 : 1 at the centre of the family), the change is not uniform:
    - 2. **The polar area (axis c):** shrinks to about **0.55×** its original area (maximum contraction!).
    - 3. **The area of the sides (leading and trailing, axis b):** shrinks to **0.80×**.
    - 4. **The area of the ends of the Earth line (axis a):** shrinks to only **0.97×** (a minimal contraction!).
    - 5. **The ratio between the polar caps (0.58×) and the equatorial belt (0.86×):** the area ratio is 0.67×, so the crater density at the poles

jumps by 1/0.67 = 1.49×!

6. **The release of gravitational energy and internal heating (§C.4):** the volume collapse and the rounding release two components of gravitational energy that turn into heat and hydrostatic pressure inside the Moon:
    - 7. **The energy of the change of shape (eccentricity):** releases **2 × 10^{27} joules** (2.1% of the binding energy, ~ 3 × 10^4 joules/kg), adding an average heating of about **30 kelvin** and keeping the core viscous and mobile.
    - 8. **The energy of the volume contraction (compression):** seven times larger — releases **1.5 × 10^{28} joules** (~ 2.1 × 10^5 joules/kg), equivalent to an average heating of **206 kelvin** (at a heat capacity of 1 kJ/kg·K).

9. **The pressure on the rigid shell and the feeding of the mare eruptions (§C.4):** forcing a stiff rocky shell (rigidity modulus $\mu \geq 10$ GPa) to change its shape by 10%–20% creates internal shear stresses of **2–12 GPa** — 20 to 120 times the shear strength of rock (~ 0.1 GPa). The shell does not shatter, but releases the hydrostatic pressure through **the feeding openings of the maria** (Paper 5), which fire the magma outward in ballistic flights and create the craters of returning material (Paper 6)!



The change of area in passing from peak flattening to the sphere at detachment: the poles contract more than any other region, so the craters on them are crowded together.

External rotational differentiation — the key statement of Paper 07 (§C.8, §D.6)

While the young Moon was attached to the Earth and turning with it, the flattening made the light material of the outer shell flow gradually **from the poles toward the equator**, leaving a thin polar shell over a window of tens of millions of years during which the craters landed evenly. In the final stage, as the body rounded up to detachment at the corotation line, the polar area converged inward, the craters were crowded together, and beneath the poles **a thin, dense crust** remained, measured today in the GRAIL data!

04 · THE MECHANISM IN WORDS

The mechanism in words: the four statistical tests and the five tests of crater-size preservation

Part D of Paper 07 puts the model of rounding and contraction to a double quantitative test: first through **four main statistical tests**, and second through **five complementary morphometric tests** that check whether the craters themselves physically shrank with the ground or were preserved whole:

1. **Test 1 — the pole/equator density ratio in the clean count:** after removing the maria and the SPA basin from both sides, the total ratio for craters of 20 km and more is **1.20 ± 0.05** (95% confidence interval: 1.10–1.30), and it rises continuously from 1.09 for craters of 20–30 km to 1.53 ± 0.19 for craters of 60–100 km and 1.84 ± 0.38 for craters of 100–300 km — **passed**.

2. **Test 2 — a change in crater diameter (a wealth of large craters at the poles):** the median diameter of the craters at the poles is **1.06** times the median diameter at the equator (95% confidence interval: 1.01–1.12) — **passed** (confirms a wealth of large, ancient craters at the poles).

3. **Test 3 — the distribution of young craters (formed after the rounding):** in two independent samples of young craters that landed after the body had already rounded into a sphere, pole/equator ratios of **0.92 and 1.12** were

measured (around 1.00, with no significant polar excess) — **matches the prediction**.

4. **Test 4 — full symmetry between the two poles:** after removing the SPA basin, which covered part of the south pole, **300 craters** per million km² were measured in the 75°–90° band at the north pole against **305** at the south pole (and in the 15° cap: 300 against 305, ratio 1.02, confidence interval 0.78–1.32) — **passed** (full equality within the counting error).

5. **The five morphometric tests of crater-diameter preservation (§D.2):** examining (a) the largest craters per equal unit of area, (b) the shape of the size distribution (the cumulative curve), (c) a comparison of ancient against fresh craters, (d) the degree of ellipticity of the crater rims, and (e) the transition diameter from simple to complex crater — **four tests established unambiguously that the crater diameters were preserved whole, and not a single test showed any sign of the craters themselves shrinking**! Had the craters shrunk with the ground, craters around 20 km would have fallen below the counting threshold and wiped out completely the measured density excess (1.20).

05 · THE KEY EQUATIONS

Academic explanation and the key equations

The differential change in the area of a unit of ground with normal $\hat{n} = (n_x, n_y, n_z)$ in passing from a triaxial ellipsoid (a, b, c) to an equivalent sphere of radius $R_s = (abc)^{1/3}$, and the gravitational binding energy W of a flattened spheroid with eccentricity e, are given in the following equations (§C.3–C.4):

$$J^{-1} = \left[n_x^2 \left(\frac{R_s}{a} \right)^2 + n_y^2 \left(\frac{R_s}{b} \right)^2 + n_z^2 \left(\frac{R_s}{c} \right)^2 \right]^{-1/2}$$

$$\frac{\rho_{\text{polar}}}{\rho_{\text{equatorial}}} = \frac{a}{c}$$

$$W = -\frac{3}{5}\frac{GM^2}{R_s}\frac{\arcsin(e)}{e}$$

$$\Delta E_{\text{compression}} \approx 1.5 \times 10^{28}\ \text{J} \quad (\Delta T \approx 206\ \text{K})$$

From the Jacobian equation $J(\lambda,\phi) = R_s^{-1}[\cos^2\lambda(a^2\cos^2\phi + b^2\sin^2\phi) + c^2\sin^2\lambda]^{1/2}$ it follows at once that at the pole ($n_z = 1$) the density is multiplied by R_s/c, at the sides ($n_y = 1$) by R_s/b, and at the ends of the Earth line ($n_x = 1$) by R_s/a — exactly the order of densities measured around the six axis ends!

06 · THE SIGNATURE

The signature in reality: the flattening vanishing before detachment and the shape kept after recession

At the end of the rounding stage, the moment the centre of mass passed the corotation line by about 210 km (r ≈ 16,610 km), the Moon detached from the Earth **already completely spherical and rigid** (maximum axis ratio ≤ 1.04, §C.1, §C.5):

- **The early flattening vanished before the detachment (not after it):** after the detachment, as the Moon receded to 30,000 km (period 14 hours) and to 60,000 km (41 hours), the tidal and rotational forces raised only a tiny elastic bulge on the rigid body (axis ratio 1.03 at 30,000 km and 1.00 at 60,000 km), and the Moon's **"fossil flattening"** today (f ≈ 0.001, frozen before 16 Earth radii) is hundreds of times smaller than the triaxial flattening of the attached stage (1.8 : 1.4 : 1).
- **Closing the circle between Papers 05, 06 and 07:** the same contraction pressure of 2–12 GPa that the rigid shell exerted during the rounding is what pushed the magma out through the feeding openings of the maria (Paper 05), fired the returning material that created the giant craters (Paper 06), and crowded them toward the two poles above a thin, dense polar crust (Paper 07)!

The lab box: the area contraction, the energy and pressure balance, and a summary of the 4 tests (Paper 07, Parts C.3–E)

Region on the Moon's face (peak flattening 1.75:1.45:1, 1,700 kg/m³)	Absolute change of area (including the drop in volume)	Displacement of the surface during the rounding	Effect on crater density and crustal thickness
The polar regions (axis c, geometric anchor)	**0.55× (0.58× in the 30° caps)**	**A stable anchor (~ +290 km polar radius)**	**Maximum density (1.49× against the equator) above a thin, dense crust**
The leading and trailing sides (axis b)	**0.80×**	Collapse inward by ~ 430 km	Intermediate density along the peripheral ring
The ends of the Earth line (axis a, sub-Earth and anti-Earth)	**0.97×**	Collapse inward by ~ 900 km	Minimal contraction — the minimum density at sub-Earth (148)
The global average of the Moon's face	**0.75×** (volume falls 1.53-fold)	Equivalent radius falls from 2,180 to 1,890 km	Compression energy 1.5 × 10^{28} J ($\Delta T \approx 206$ K)

Empirical test (Part D)	Definition and sample	Measured result	Status
Test 1: pole/equator density ratio	**Clean count (without maria and SPA), 30° caps against** $\lvert\phi\rvert < 15^\circ$	**1.20 ± 0.05 (1.09 at 20–30 km → 1.53 at 60–100 km)**	**Passed**
Test 2: median crater diameter	**Comparing crater sizes at the poles against the equator**	**The polar median is 1.06 times larger (confidence interval 1.01–1.12)**	**Passed**
Test 3: young craters (after the rounding)	Two independent samples of late craters	Ratios of **0.92 and 1.12** (no polar excess)	**Matches the prediction**
Test 4: symmetry between the two poles	**The 75°–90° band and the 15° cap (without SPA)**	**300 in the north against 305 in the south per million km² (1.02)**	**Passed**
5 tests of crater-size preservation	Largest, size distribution, ancient/fresh, ellipticity, transition to complex	**4 of 5** confirm full preservation, **0** show crater shrinkage	**Passed**

The door ahead

In Chapters 13–14 we solved the riddle of the Moon's poles (*Why the Poles Are Crowded*): we discovered how the passage from a flattened triaxial body (1.8 : 1.4 : 1) to a rigid sphere (3,778 km) compressed the poles by 1.20–1.53 times above a thin polar crust that had undergone "external rotational differentiation", while fully preserving the original crater diameters.

But in Chapter 13 an unexpected discovery appeared at the heart of the south pole: **the South Pole–Aitken basin (SPA)** — the largest and most mysterious basin on the Moon (2,400 km), at whose core (0°–10°) the crater density drops to 132 per million km²! What is the SPA basin really in the LEF model, why is its shape elliptical rather than round, and how did it form? In the next chapter — **Chapter 15: The South Pole–Aitken Basin (SPA) — a Detachment Wound or the Largest Impact Crater?** (from Paper 08 — *South Pole–Aitken*) — we will dive into this giant basin!

Chapter 15

The South Pole–Aitken Basin (SPA) — a Detachment Wound or the Largest Impact Crater? →

← Chapter 13 · Prologue

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